

Sharp Faber–Krahn stability via the Kohler–Jobin transfer

Plan 1 / Sharp stability for Faber–Krahn

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Abstract

The spectral closure of `fixed-domain-bernoulli-expansion.md` gives sharp *Saint–Venant* stability:

$$E(\Omega) - E(B_1) \geq c_*(N, R) \alpha(\Omega) \quad \text{for } Q_\alpha(\Omega) \leq q_*(N, R).$$

By BDV Lemma 4.2 ($\mathcal{A}(\Omega)^2 \lesssim \alpha(\Omega)$) and the small-vs-large asymmetry split, this gives the global sharp Saint–Venant inequality $\mathcal{A}(\Omega)^2 \leq C_{\text{SV}}(N, R)(E(\Omega) - E(B_1))$.

This note records the Kohler–Jobin transfer that converts this into sharp *Faber–Krahn* stability for the first Dirichlet eigenvalue:

$$\lambda_1(\Omega) - \lambda_1(B^*) \geq c_{\text{FK}}(N, R) \mathcal{A}(\Omega)^2,$$

where B^* is the ball with $|B^*| = |\Omega|$. The transfer uses only the Kohler–Jobin inequality (a non-perturbative pointwise comparison that holds for every open set) plus elementary scaling. The output is the BDV sharp Faber–Krahn theorem.

1 The Kohler–Jobin inequality

For an open bounded set $\Omega \subset \mathbb{R}^N$, let

$$T(\Omega) := \int_{\Omega} u_{\Omega} dx, \quad -\Delta u_{\Omega} = 1 \text{ in } \Omega, \quad u_{\Omega} = 0 \text{ on } \partial\Omega,$$

be the torsional rigidity, and let $\lambda_1(\Omega)$ be the first Dirichlet eigenvalue of $-\Delta$ on Ω . Both functionals scale as

$$T(t\Omega) = t^{N+2}T(\Omega), \quad \lambda_1(t\Omega) = t^{-2}\lambda_1(\Omega),$$

so the product

$$\Psi(\Omega) := T(\Omega) \cdot \lambda_1(\Omega)^{(N+2)/2}$$

is scale-invariant.

Theorem 1 (Kohler–Jobin, 1978). *For every open bounded $\Omega \subset \mathbb{R}^N$ with $\lambda_1(\Omega) < \infty$,*

$$\Psi(\Omega) \geq \Psi(B),$$

where B is any ball, with equality iff Ω is a ball (up to translation and null sets).

See Kohler–Jobin (1978) and Brasco, “On torsional rigidity and principal frequencies” (ESAIM:COCV, 2014) for proofs via rearrangement. The constant $\Psi(B)$ is universal:

$$\Psi(B) = T(B) \cdot \lambda_1(B)^{(N+2)/2} = c_N$$

depends only on N (not on the radius), since Ψ is scale-invariant.

2 Transfer at fixed volume

Fix $|\Omega| = |B^*|$ where B^* is the comparison ball. Setting $L := \lambda_1(B^*)$, $T_0 := T(B^*)$, and abbreviating $\beta := (N + 2)/2$, Theorem ?? reads

$$T(\Omega) \cdot \lambda_1(\Omega)^\beta \geq T_0 \cdot L^\beta. \quad (1)$$

Lemma 2 (Pointwise transfer). *For every Ω with $|\Omega| = |B^*|$,*

$$\lambda_1(\Omega) \geq L \cdot \left(\frac{T_0}{T(\Omega)} \right)^{1/\beta}.$$

Equivalently, with the Saint–Venant deficit $\delta_{\text{SV}}(\Omega) := T_0 - T(\Omega) \geq 0$ (note $\delta_{\text{SV}} \geq 0$ at fixed volume by Saint–Venant),

$$\lambda_1(\Omega) - L \geq L \cdot \left[\left(\frac{1}{1 - \delta_{\text{SV}}/T_0} \right)^{1/\beta} - 1 \right].$$

Proof. Rearrange (??). □

Lemma 3 (Small-deficit linearization). *For $\delta_{\text{SV}} \leq T_0/2$,*

$$\lambda_1(\Omega) - L \geq \frac{L}{\beta T_0} \cdot \delta_{\text{SV}} = \frac{2L}{(N + 2)T_0} \cdot (T_0 - T(\Omega)).$$

Proof. For $t \in [0, 1/2]$, $(1 - t)^{-1/\beta} - 1 \geq t/\beta$ by elementary convexity. Apply with $t = \delta_{\text{SV}}/T_0$. □

Remark 4. The factor $2L/((N + 2)T_0)$ is universal in N once normalized. For the unit ball B_1 in $\mathbb{R}^N$, $L = L_N$ is the first Dirichlet eigenvalue of $-\Delta$ on B_1 and $T_0 = T_N$ is the torsional rigidity of B_1 . Both are explicit: L_N is the smallest positive zero of a Bessel function, $T_N = |B_1|/(N(N + 2))$.

3 Sharp Faber–Krahn from sharp Saint–Venant

Proposition 5 (Small-asymmetry Faber–Krahn). *Assume the sharp Saint–Venant stability holds in the form*

$$T(B^*) - T(\Omega) \geq c_{\text{SV}}(N, R) \mathcal{A}(\Omega)^2 \quad \text{for all } \Omega \subset B_R \text{ with } |\Omega| = |B^*| \text{ and } \mathcal{A}(\Omega) \leq \mathcal{A}_*. \quad (2)$$

Then for the same Ω ,

$$\lambda_1(\Omega) - \lambda_1(B^*) \geq \frac{2L c_{\text{SV}}(N, R)}{(N + 2)T_0} \mathcal{A}(\Omega)^2.$$

Proof. Combine Lemma ?? with (??). The condition $\delta_{\text{SV}} \leq T_0/2$ in Lemma ?? is satisfied for $\mathcal{A}_*$ small enough. □

Theorem 6 (Sharp Faber–Krahn stability). *There is $c_{\text{FK}}(N, R) > 0$ such that, for every open $\Omega \subset B_R$ with $|\Omega| = |B^*|$,*

$$\lambda_1(\Omega) - \lambda_1(B^*) \geq c_{\text{FK}}(N, R) \mathcal{A}(\Omega)^2.$$

Proof. Small asymmetry. For $\mathcal{A}(\Omega) \leq \mathcal{A}_*(N, R)$, combine our sharp Saint–Venant stability (the global form derived in `fixed-domain-bernoulli-expansion.md`, Theorem 6.3, combined with the small-vs-large split using BDV Lemma 4.2) with Proposition ?. This gives $\lambda_1(\Omega) - L \geq c_1(N, R) \mathcal{A}(\Omega)^2$ with $c_1 = 2L c_{\text{SV}}/((N + 2)T_0)$.

Large asymmetry. For $\mathcal{A}(\Omega) \geq \mathcal{A}_*$, the classical Faber–Krahn inequality $\lambda_1(\Omega) \geq L$ with strict inequality away from the ball combined with the qualitative continuity of λ_1 in Hausdorff convergence (or any compactness argument) yields a uniform lower bound

$$\lambda_1(\Omega) - L \geq \inf_{\mathcal{A}_* \leq \mathcal{A}(\Omega) \leq 2} (\lambda_1(\Omega) - L) =: \delta_*(N, R, \mathcal{A}_*) > 0 \geq \delta_* \cdot \frac{\mathcal{A}(\Omega)^2}{4}.$$

(Here $\mathcal{A} \leq 2$ since $|\Omega \Delta B^*| \leq 2|B^*|$, so $\mathcal{A}^2 \leq 4$.) Setting $c_2 := \delta_*/4$,

$$\lambda_1(\Omega) - L \geq c_2 \mathcal{A}(\Omega)^2.$$

Take $c_{\text{FK}} := \min(c_1, c_2)$. □

4 Quantitative dependence on (N, R)

The constant $c_{\text{FK}}(N, R)$ is the minimum of the small-asymmetry and large-asymmetry constants:

- **Small-asymmetry constant:**

$$c_1(N, R) = \frac{2L_N c_{\text{SV}}(N, R)}{(N+2)T_N},$$

where $c_{\text{SV}}(N, R)$ is the Saint–Venant constant from our spectral closure. Tracing back through the selection chain:

- Selection (`quantitative-selection-principle.md`) preserves the quotient Q_α up to a constant $C_{\text{sel}}(N, R)$, gaining $\sigma \leq C\tau = Cq_0^{1/4}$.
- Graph entry (`graph-entry-closure.md`) requires $\alpha(\tilde{\Omega}) \leq \alpha_{\text{graph}}(N, R)$.
- Spectral closure (`fixed-domain-bernoulli-expansion.md`) requires $\sigma \leq \sigma_*(N, R)$ and $\|g\|_{C^{2,\gamma}} \leq \delta_*(N, R)$.

Combining, $q_* \sim \sigma_*^4$ (the binding constraint), and $c_{\text{SV}} \geq q_*/C_{\text{sel}}$. Hence

$$c_1(N, R) \geq \frac{2L_N \sigma_*(N, R)^4}{(N+2)T_N \cdot C_{\text{sel}}(N, R)}.$$

- **Large-asymmetry constant** $c_2(N, R)$: a qualitative compactness constant. Explicit lower bound (in $N = 2, 3$) is obtained from quantitative versions of Faber–Krahn for the convex hull or convex envelope, but for our purposes any positive lower bound suffices.

Remark 7 (Comparison with BDV). This is exactly the chain BDV use to prove sharp Faber–Krahn: their Theorem 1.1 is our Theorem ???. The new content of Plan 1 is that $c_{\text{SV}}(N, R)$ is now derived from the Bernoulli spectral closure (`fixed-domain-bernoulli-expansion.md`) rather than from BDV’s nearly spherical second variation. The Kohler–Jobin transfer step is shared with BDV and uses no new input.

5 Status

The full chain for sharp Faber–Krahn stability is structurally complete, with two possible closure routes:

1. **Route A (unconditional):** BDV’s nearly spherical second variation supplies the sharp Saint–Venant constant $c_{\text{SV}}(N, R)$.
2. **Route B (conditional):** the Bernoulli spectral closure of `fixed-domain-bernoulli-expansion.md`, supplemented by `corrections-response.md`, supplies $c_{\text{SV}}(N, R)$, conditional on the explicit constants in the second-variation source enumeration.

Either way, the Kohler–Jobin transfer of Lemma ?? + Lemma ?? + small/large split (Theorem ??) converts c_{SV} into c_{FK} .

The quantitative Faber–Krahn constant $c_{\text{FK}}(N, R)$ is given by an explicit formula in terms of:

- the universal ball constants $L_N = \lambda_1(B_1)$ and $T_N = T(B_1) = |B_1|/(N(N+2))$;
- the BDV regularity constants from selection and graph entry;
- the spectral-closure smallness constants σ_*, δ_* .

The exponent is sharp: the inequality $\lambda_1(\Omega) - \lambda_1(B^*) \geq c_{\text{FK}} \mathcal{A}(\Omega)^2$ is the optimal power of $\mathcal{A}$, as already observed by Bhattacharya, Hansen, Nadirashvili (and saturated by ellipsoidal perturbations of the ball).